



COURSE DESCRIPTION FORM: CS-3001 Computer Networks

FAST School of Computing, National University of Computer and Emerging Sciences,
Islamabad

BSCS – Fall 2025

PROGRAM TO BE EVALUATED

Course Description

Course Code	CS-3001																							
Course Title	Computer Networks																							
Credit Hours	3+1																							
Prerequisites by Course(s) and Topics	CS-2001 Data Structures																							
Grading Policy	Absolute grading																							
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. For a missed sessional / final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.																							
Course Plagiarism Policy	Plagiarism in project or sessional/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.																							
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	<table><tr><td colspan="3">Assessment Items</td></tr><tr><td>Assessment Item</td><td>Number</td><td>Weight (%)</td></tr><tr><td>Assignments</td><td>4</td><td>10</td></tr><tr><td>Project (Lab)</td><td>1</td><td>10</td></tr><tr><td>Sessional Exams</td><td>2</td><td>30 (15 each)</td></tr><tr><td>Quizzes</td><td>4</td><td>10</td></tr><tr><td>Final Exam</td><td>1</td><td>40</td></tr></table>			Assessment Items			Assessment Item	Number	Weight (%)	Assignments	4	10	Project (Lab)	1	10	Sessional Exams	2	30 (15 each)	Quizzes	4	10	Final Exam	1	40
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Course Instructors	Dr. Ali Zeeshan, Ms. Sana Razzaq, Mr Furqan																							
Lab Instructors (if any)																								
Course Coordinator																								
URL (if any)																								
Current Catalog Description	Basic computer and communication networking technologies, concepts of layering, routing, congestion control, and medium access control. The course primarily aims to acquaint the student with basic computer and communication networking technologies and the layered approach that makes design, implementation and																							



	operation of computer and communication networks possible. The course explores in detail the concepts of layering, routing, congestion control, and medium access control. In keeping with the nature of the textbook, the course follows a top-down approach, commencing with networking applications to motivate the study of networking protocols, architectures and principles.									
Textbook (or Laboratory Manual for Laboratory Courses)	James F. Kurose and Keith W. Ross, "Computer Networking: A Top-Down Approach", 8 th Edition, Pearson.									
Reference Material	1. Behrouz A. Forouzan, "Data Communications and Networking" 5th Edition 2. Andrew S. Tanenbaum and David J. Wetherall, "Computer Networks" 5th Edition, Pearson. 3. William Stallings, "Data and Computer Communications" 8th Edition, Pearson.									
Course Learning Outcomes	A. Course Learning Outcomes (CLOs) After completion of the course, the students shall be able to: 1. Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack. 2. Demonstrate the basics of network concepts using state-of-the-art network tools. 3. Demonstrate operations of various classical routing and switching protocol via simulators. 4. Apply Socket Programming (client/server) to solve various real world problems, including ensuring of data integrity B. Program Learning Outcomes <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%; text-align: center;">1</td><td style="width: 40%;">Academic Education</td><td>Completion of an accredited program of study designed to prepare graduates as computing professionals</td></tr> <tr> <td style="text-align: center;">2</td><td>Knowledge for Solving Computing Problems</td><td>Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements</td></tr> <tr> <td style="text-align: center;">3</td><td>Problem Analysis</td><td>Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines</td></tr> </table>	1	Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals	2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements	3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
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	4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations									
	5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations									
	6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings									
	7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions									
	8	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice									
	9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice									
	10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional									
	C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)											
			PLOs									
			1	2	3	4	5	6	7	8	9	10
CLOs	1	✓	✓									
	2		✓	✓		✓						
	3			✓	✓	✓						
	4			✓	✓							

Topics covered in the course with number of lectures on each topic (assume 15 weeks of instruction and 1.5 hour lecture duration)	Week No	List of Topics	Assessment	CLO(s)
	1	Computer Networks and Internet: Internet and Protocols Internet structure Network edge, Access Network, Network core Physical media Types		1
	2	Protocol layers, service models Network Edge, Network Core, Delays and Service Models, Networks Security	Assignment 1	1
	3	Application Layer: Principles and Protocols Web & HTTP, Cookies and Web Caches	Quiz 1	1,2
	4	FTP, E-mail, SMTP, IMAP Domain Name System (DNS)		1,2
	5	Socket Programming with UDP and TCP Revision	Sessional-I	4
	6	Transport Layer: Services Multiplexing and demultiplexing Connection Less Transport: UDP	Assignment 2	1,2
	7	Principles of Reliable Data Transfer Pipelining Protocols, Go Back N, Selective Repeat		1,3
	8	Connection-oriented transport: TCP TCP Segment Structure TCP RTT Estimation and Timeout TCP Flow Control	Quiz 2	1,2
	9	Principles of congestion control TCP congestion control Evolution of Transport-Layer Functionality QUIC	Assignment 3	1
	10	Network layer: Data plane overview IP: the Internet Protocol Datagram format IP addressing, Subnetting	Quiz 3	1
	11	VLSM Network address translation (NAT) DHCP	Sessional-II	1,2
	12	IPv6 vs IPv4 and Transition Generalized Forwarding SDN Middleboxes	Project	1,2
	13	Network Layer: Control Plane Routing Algorithms Link State Routing Distance Vector Routing		3
	14	RIP, OSPF and BGP The SDN Control Plane	Assignment 4	3



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	15	Link Layer and LANs Introduction to the Link Layer Error-Detection and –Correction Techniques Multiple Access Links and Protocols Switched LANs, VLANs	Quiz 4	1,2,3
	16	Physical Layer Wireless and Mobile Networks		1,2
Laboratory Projects/ Experiments Done in the Course	Lab outline is separately provided			
Class Time Spent (in percentage)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)
	50	25	20	5
Oral and Written Communications	Every student is required to submit at least __4__ written reports of typically _5__ pages and to make __1__ oral presentation of typically ____10__ minutes' duration.			